

Week 2 Progress Report

Literature Review (continued) and Dataset Collection

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

Student: Jayanth Akarapu

Course: Open Data Practice - Research Practicum

Supervisor: Dr. Prasanth Nayak

Gantt activities this week: Literature Review (continued), Dataset Collection

Section	Content
1	Week 2 scope against the project Gantt chart
2	Literature review: thematic synthesis, method comparison matrix, refined gaps
3	Dataset collection: selection, acquisition, data dictionary, quality audit, first look
4	Ethics and data governance
5	Progress status, risks and Week 3 plan

1. Week 2 Scope

The project Gantt chart (CA1 proposal, Section 3.8) runs Literature Review from August to the end of September and Dataset Collection across August. Data Preprocessing begins in September. Week 2 therefore has two tasks: move the literature review from paper-by-paper summaries to a critical synthesis, and obtain, document and audit the dataset that the experiments will use.

Gantt activity	Planned window	Week 2 target	Status
Literature Review	Aug - end Sep	Thematic synthesis, comparison matrix, refined gaps	On track
Dataset Collection	Aug	Dataset selected, downloaded, verified and profiled	Complete
Data Preprocessing	Sep - end Oct	Not started; issues logged in Section 3.6 feed it	Next

2. Literature Review (continued)

2.1 Thematic synthesis

Week 1 summarised 15 papers (2025-2026) individually. This week the papers were read against each other to find where they agree, where they disagree and what none of them tests. Five findings shape the study design.

(a) Tree ensembles are the consistent winner on tabular security data. Random Forest is the best or joint-best model in Papers 4, 10, 11 and 12, and the IDS survey (Paper 7) reports the same across NSL-KDD, UNSW-NB15 and CICIDS2017. SVM is competitive but costs more compute (Paper 12). This supports the proposal's choice of Decision Tree, Random Forest and SVM, with Random Forest as the expected leader and Decision Tree as the interpretable reference.

(b) Reported scores are high and hard to compare. Accuracy values of 96-99% (Papers 4, 9, 10, 11) come from different datasets, splits and imbalance treatments. Only Paper 10 states its resampling method (SMOTE). None reports a simple rule-based comparator, so it is unclear how much the ML adds over the rules banks already run. This study will therefore report precision, recall, F1 and ROC-AUC on one fixed held-out split, and include a rule baseline.

(c) Big data architecture is described more than it is measured. Papers 1 and 2 argue for Kafka/Spark/Flink pipelines, and Paper 1 reports sub-second latency, but on simulated data and without a model-by-model cost comparison. Measuring training and scoring time as data size grows gives a practical, testable answer to RQ1 on the scale this project can run.

(d) Robustness and drift are named as open problems. Paper 7 lists concept drift and adversarial robustness as unresolved; Paper 6 shows adversarial training helps but only on synthetic attack scenarios. A time-ordered split (train on earlier months, test on later ones) and a test on entities unseen in training are cheap ways to probe this within scope.

(e) Governance is the Irish and EU angle. Papers 8, 13, 14 and 15 connect detection to GDPR, federated learning, DORA/NIS2 and the EU AI Act. For Irish banks, an explainable model is a compliance need, not an extra. Feature importance (and SHAP where feasible) will be reported to answer RQ3 and RQ4.

2.2 Method comparison matrix

#	Study	Models	Data	Class balance	Metrics / result	Use here
1*	Sathupadi et al. 2025	Kafka + ML classifiers	Simulated banking	n/s	Sub-second latency	Pipeline design
2	Kumar et al. 2025	RF, SVM, NN (review)	Security datasets	n/s	Up to 40% over rules	Big data framing
3	Kokogho et al. 2025	Predictive risk scoring	Fintech transactions	n/s	+35% early detection	Risk context
4*	Ali et al. 2025	RF, SVM, DT, GB	CICIDS2017	n/s	RF Acc 98.4%, F1 0.97	Evaluation protocol

#	Study	Models	Data	Class balance	Metrics / result	Use here
5	Mizanur et al. 2025	IF, AE, K-Means + RF, SVM, NB	Network traffic	Analyse d	P 0.96, R 0.94	Anomaly option
6	Padmanaban et al. 2026	RF, SVM + adversarial training	Synthetic banking attacks	n/s	+42% robustness	Robustness
7*	Hozouri et al. 2025	Survey, 150+ IDS papers	NSL-KDD, UNSW-NB15, CICIDS2017	Discuss ed	RF, CNN-LSTM best	Theory; model choice
8	Laddi et al. 2025	Federated IDS	Network nodes	n/s	95.6% Acc; within 2-3% of central	GDPR angle
9	Zhang 2025	CNN-LSTM	CIC-IoT2023	n/s	99.1% Acc, FPR < 0.4%	DL contrast
10*	Naeem et al. 2026	DT, RF, SVM	Digital banking	SMOTE	RF AUC 0.987	Benchmark to beat
11	Dhammayogi & Pramatha 2026	RF (50-500 trees)	Banking transactions	n/s	Acc 98.7% at 200 trees	Tuning range
12	Sanda et al. 2026	DT, RF, SVM, LR, KNN, NB	Banking fraud	n/s	RF best overall	Extra baselines
13	Eshra et al. 2025	CTI + ML prioritisation	SOC simulation	n/a	MTTD -58%	Prevention side
14	Ullah et al. 2026	Predictive cyber-risk	Financial + cyber events	n/a	Losses -31%	Risk quantification
15	Kovacevic et al. 2025	Review of AI tools	Banking cases	n/a	Fraud losses -52%	Governance risks

* baseline paper. n/s = not stated in the paper; n/a = not a classification study.

2.3 Refined research gaps and link to research questions

Gap from the synthesis	How this study responds	RQ
G1. High scores are not compared with the simple rules banks already use	Add a transparent rule baseline next to DT, RF and SVM on the same split	RQ2
G2. Results mix splits and imbalance methods, so they cannot be compared	One fixed stratified 80/20 split and seed; imbalance treatment stated and compared	RQ2
G3. Big data claims lack cost measurements	Record training and scoring time as sample size grows	RQ1, RQ3
G4. Drift and generalisation are named but rarely tested	Time-ordered split and a held-out-entity test	RQ2, RQ3
G5. Little Irish/EU regulatory grounding of the models	Explainability outputs and GDPR / AI Act discussion in recommendations	RQ3, RQ4

2.4 Datasets used in the literature

Most reviewed studies use network-intrusion benchmarks (CICIDS2017, NSL-KDD, UNSW-NB15, CIC-IoT2023) or unnamed banking datasets. Network-flow data does not describe banking transactions, and unnamed datasets cannot be reproduced. This pushed the selection in Section 3 toward an open, labelled, transaction-level banking dataset.

3. Dataset Collection

3.1 Selection criteria and candidates

Five criteria were set before choosing: (1) transaction-level banking records, (2) a fraud or threat label, (3) an open licence permitting academic use, (4) a size that trains the three models on a laptop in minutes, and (5) no personal data. The Week 1 plan also listed the Bank Account Fraud (BAF) dataset, so it was assessed against the same criteria.

Candidate	Type	Size	Label	Licence	Decision
Bank Transaction Records along Suspicious Flags (Kaggle)	Simulated bank account transactions	33,611 rows	isSuspicious	MIT	Selected: named in the CA1 proposal, meets all five criteria
Bank Account Fraud suite (Jesus et al., NeurIPS 2022)	Synthetic account-opening applications	1M rows per variant	fraud_bool	CC BY-NC-SA 4.0	Reserve: application fraud, not transactions; large for SVM
PaySim (Lopez-Rojas et al., 2016)	Simulated mobile-money transfers	6.36M rows	isFraud	CC BY-SA 4.0	Reserve: mobile money rather than bank accounts
CICIDS2017	Network flows	about 2.8M flows	Attack type	Academic use	Rejected: network traffic, not banking transactions

3.2 Acquisition and verification

The dataset was downloaded with the Kaggle API: `kaggle datasets download -d charanmaik/bank-transaction-records-along-suspicious-flags --unzip -p data`. The API token is kept in `~/ .kaggle/` and is not committed. Kaggle metadata lists the licence as MIT and describes the data as a realistic simulation of bank transactions. The file `transactions_Dataset.csv` is 2.13 MB; its SHA-256 checksum is recorded so later weeks can prove the same file is used:

3863ecc55dd883f959c79362de9a2425ad8ac0ec05e5d6d71890488fc65d83f1

3.3 Data dictionary

Column	Type	Missing	Unique	Min	Max	Meaning
date	object	0	757			Transaction date (dd-mm-yyyy string, parsed to datetime in preprocessing)
description	object	0	43			Channel / merchant code, e.g. NETTXN:EBAY (categorical text)
debit	float64	0	18,521	0.0	1793799.36	Amount leaving the account (0 when the row is a credit)
credit	float64	0	15,083	0.0	1993110.41	Amount entering the account (0 when the row is a debit)
balance	float64	0	33,611	-720071.17	22218339.63	Account balance after the transaction
isSuspicious	int64	0	2	0.0	1.0	Target label: 1 = flagged suspicious, 0 = normal

3.4 Initial quality audit

Check	Result	Implication
Rows x columns	33,611 x 6	Small enough to run SVM without sampling
Missing cells	0	No imputation needed
Exact duplicate rows	0	No de-duplication needed

Check	Result	Implication
Date parsing (dd-mm-yyyy)	0 failures; 2022-12-16 to 2025-01-12 (757 distinct days)	Enables a time-ordered split and date features
Negative values	debit/credit: 0; balance: 310	Negative balances read as overdrafts; keep
Rows with both debit and credit > 0	0	Each row is one direction
Rows with neither debit nor credit	0	Every row moves money
Balance values unique	True	Balance acts like a row ID; leakage risk
Balance = previous balance + credit - debit	2.15% of rows	Rows are not one running account; balance is not a ledger
Distinct descriptions	43	Low-cardinality categorical; one-hot encode

3.5 Class distribution

14,203 of 33,611 transactions (42.3%) are flagged suspicious, a 1.37:1 ratio of normal to suspicious. This is only mildly imbalanced, far from card-fraud benchmarks (about 578:1), and much higher than a real bank's fraud rate. Two consequences: class weights should be enough and SMOTE is tested only as a comparison, and results cannot be read as real-world alert volumes. Precision, recall and F1 on the suspicious class stay the main metrics.

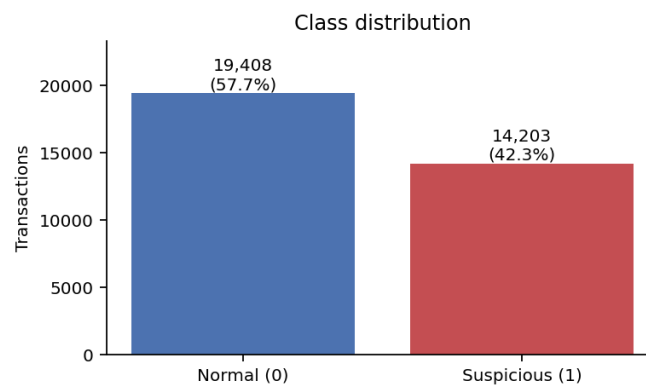


Figure 1. Class distribution of isSuspicious.

3.6 First look at the data

The suspicious rate differs sharply by description (Figure 2). 26 of the 43 descriptions are never flagged, while some channels are flagged most of the time. The description field alone is therefore highly predictive. That is useful, but it also warns that models may learn the labelling rule rather than general threat behaviour. A test on descriptions held out of training will be added in model evaluation to check this. Descriptions split into two groups: payment channels such as PCA, IMPS and cash deposit are never flagged, while online merchants, ticket booking and crypto exchanges are flagged in roughly 70-75% of rows.

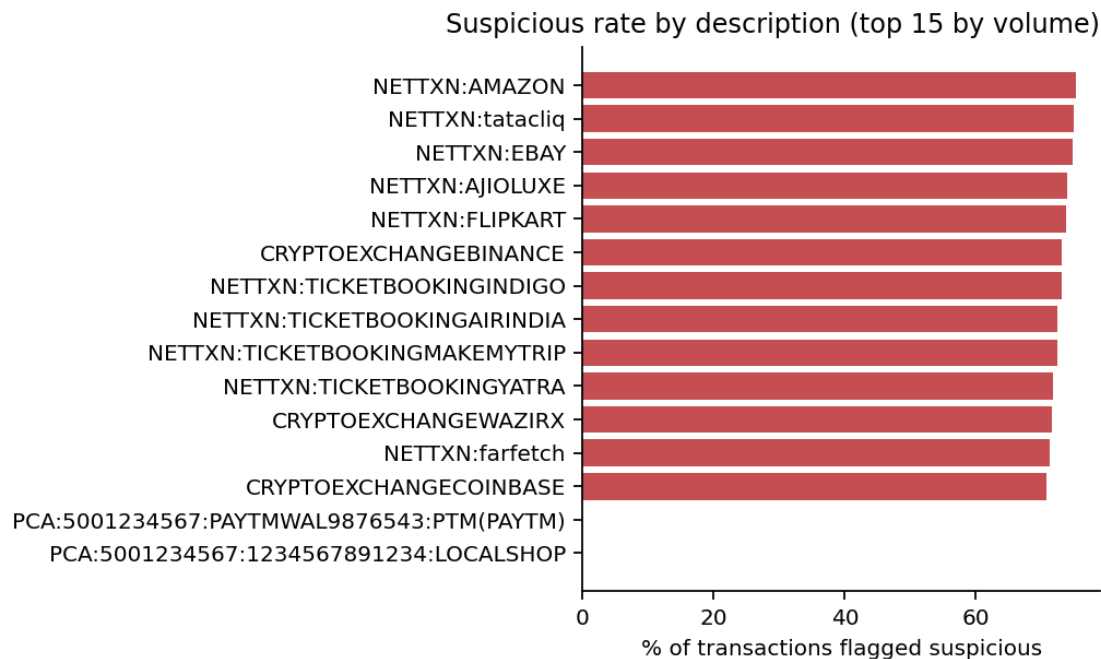


Figure 2. Share of suspicious transactions for the 15 most frequent descriptions.

Suspicious transactions are larger. Among debit rows the median is 571,350 for suspicious against 115,298 for normal; among credit rows it is 596,527 against 416,824 (Figure 4). Amounts span several orders of magnitude, so a log transform or robust scaling will be used. Over time the data is not uniform (Figure 3): a typical month has about 724 rows and a suspicious rate near 20%, but 2023-03 holds 10,637 rows with a 60% rate. A random split would mix this spike into both train and test, so a time-ordered split will be reported alongside it.

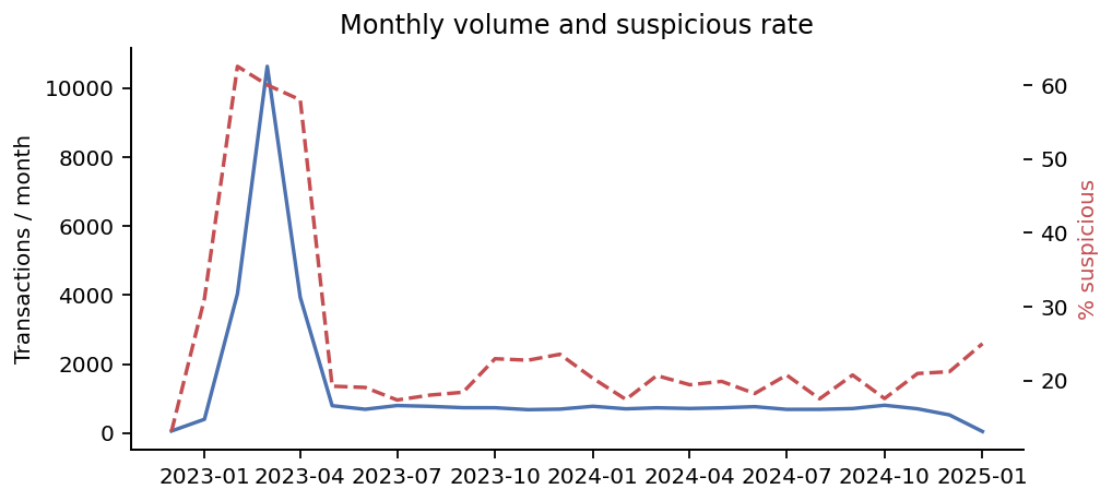


Figure 3. Monthly transaction volume (solid) and suspicious rate (dashed).

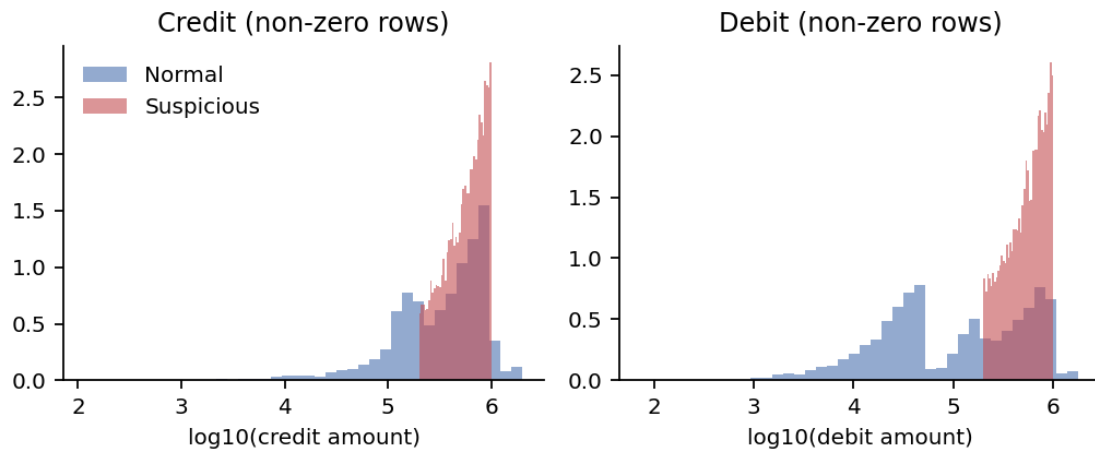


Figure 4. Distribution of log10 credit and debit amounts by class.

Issues logged for Data Preprocessing (Week 3 onward)

#	Issue	Planned action
1	date is text	Parse; derive month, day of week, day of month
2	description is text with a channel prefix (e.g. NETTXN:)	Split prefix and merchant; one-hot encode
3	Amounts are highly skewed	log1p transform; RobustScaler for SVM
4	balance is unique per row	Test with and without balance to avoid leakage
5	1.37:1 class ratio	Stratified split; class weights by default, SMOTE on training data only as a comparison
6	Volume spike in 2023-03	Report a time-ordered split next to the random split
7	Description may encode the label	Add held-out-description and rule-baseline checks

4. Ethics and Data Governance

The dataset is simulated and contains no names, account numbers or other personal data, so GDPR and the Irish Data Protection Act 2018 are not engaged by the raw data. It is published under the MIT licence, which permits academic use with attribution. Account-style numbers inside the description field (e.g. PCA:5001234567:...) are placeholder values, not real identifiers, and are removed when the prefix is split out. The file is stored locally, is excluded from version control by .gitignore, and is identified by checksum. Because the data is simulated and its merchant names are not Irish, results will be presented as evidence about methods, not as measurements of Irish banks.

5. Progress, Risks and Week 3 Plan

Item	Status / note
Literature synthesis, comparison matrix, refined gaps	Done (Section 2)
Dataset selected against written criteria	Done (Section 3.1)
Dataset downloaded, checksummed, licence checked	Done (Section 3.2)
Data dictionary and quality audit	Done; saved as data_dictionary.csv and dataset_profile.json
Risk: labels may follow a simple rule	Mitigated by a rule baseline and held-out-description test
Risk: simulated data limits the Irish claim	Stated as a limitation; Central Bank of Ireland statistics used for context

Week 3 (Literature Review continues; Data Preprocessing starts): add 3-5 papers on concept drift and rule-versus-ML comparisons; start the preprocessing notebook (date parsing, description split, log amounts, encoding); create the stratified 80/20 split with a fixed seed and save it for all later experiments.

References

1. Ali, M., Raza, A., Akram, M.A., Arif, H. and Ali, A. (2025) 'Machine Learning-Driven Approaches to Cyber Threat Detection: Enhancing IoT Security', Journal of Informatics and IT.
2. Dhammayogi, M.B.D. and Pramatha, C.R.A. (2026) 'Early Detection of Digital Transaction Fraud in Banking Systems Using the Random Forest Algorithm', JITECS.
3. Hozouri, A., Mirzaei, A. and Effatparvar, M. (2025) 'A Comprehensive Survey on Intrusion Detection Systems with Advances in ML, DL and Emerging Cybersecurity Challenges', Discover Artificial Intelligence.
4. Jesus, S., Pombal, J., Alves, D., Cruz, A., Saleiro, P., Ribeiro, R., Gama, J. and Bizarro, P. (2022) 'Turning the Tables: Biased, Imbalanced, Dynamic Tabular Datasets for ML Evaluation', NeurIPS Datasets and Benchmarks Track.
5. kaggle.com (2025) Bank Transaction Records along Suspicious Flags. Available at: <https://www.kaggle.com/datasets/charanmaik/bank-transaction-records-along-suspicious-flags>
6. Kovacevic, A., Radenkovic, S. et al. (2025) 'Artificial Intelligence and Cybersecurity in the Banking Sector: Opportunities and Risks', Contemporary Issues in Economics and Business.
7. Kumar, B.H., Nuka, S.T., Malempati, M. et al. (2025) 'Big Data in Cybersecurity: Enhancing Threat Detection with AI and ML', Metallurgical and Materials Engineering.
8. Lopez-Rojas, E.A., Elmir, A. and Axelsson, S. (2016) 'PaySim: A Financial Mobile Money Simulator for Fraud Detection', 28th European Modeling and Simulation Symposium.
9. Naeem, W., Butt, M.A. and Javeid, U. (2026) 'Machine Learning-Based Fraud Detection Systems and Their Effectiveness in Reducing Cybersecurity Risks in Digital Banking', Social Science Review Archives.
10. Padmanaban, H., Sharma, Y.K., Sharma, P. et al. (2026) 'Adversarial Machine Learning Framework for Robust Banking Security: The ACTP Tool', Journal of The Institution of Engineers (India).
11. Sanda, A.M., Mukhtar, U.S. et al. (2026) 'Enhanced Banking Fraud Detection: A Comparative Analysis of Supervised Machine Learning Algorithms', FETICON Proceedings.
12. Sathupadi, K., Achar, S., Bhaskaran, S.V. et al. (2025) 'BankNet: Real-Time Big Data Analytics for Secure Internet Banking', Big Data and Cognitive Computing, 9(2), p. 24.
13. Sharafaldin, I., Lashkari, A.H. and Ghorbani, A.A. (2018) 'Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization', ICISSP.
14. Full list of the 15 reviewed papers: Week 1 Literature Review (v2), references 1-15.